

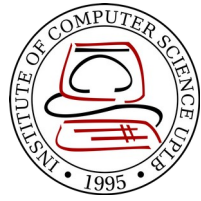
CMSC 137

Data Communications and Networking

Lecture Module
11 - Switching



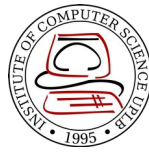
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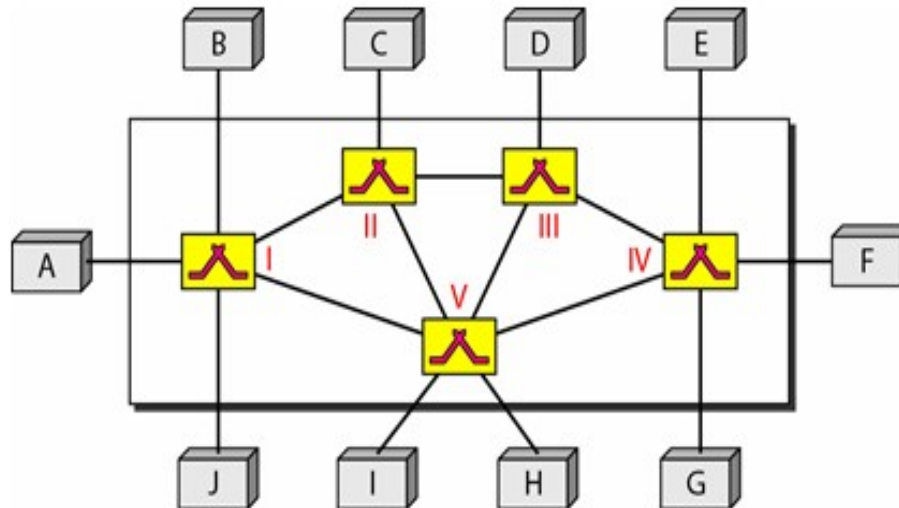
Module 11: Switching

Rozano S. Maniaol
Lecturer

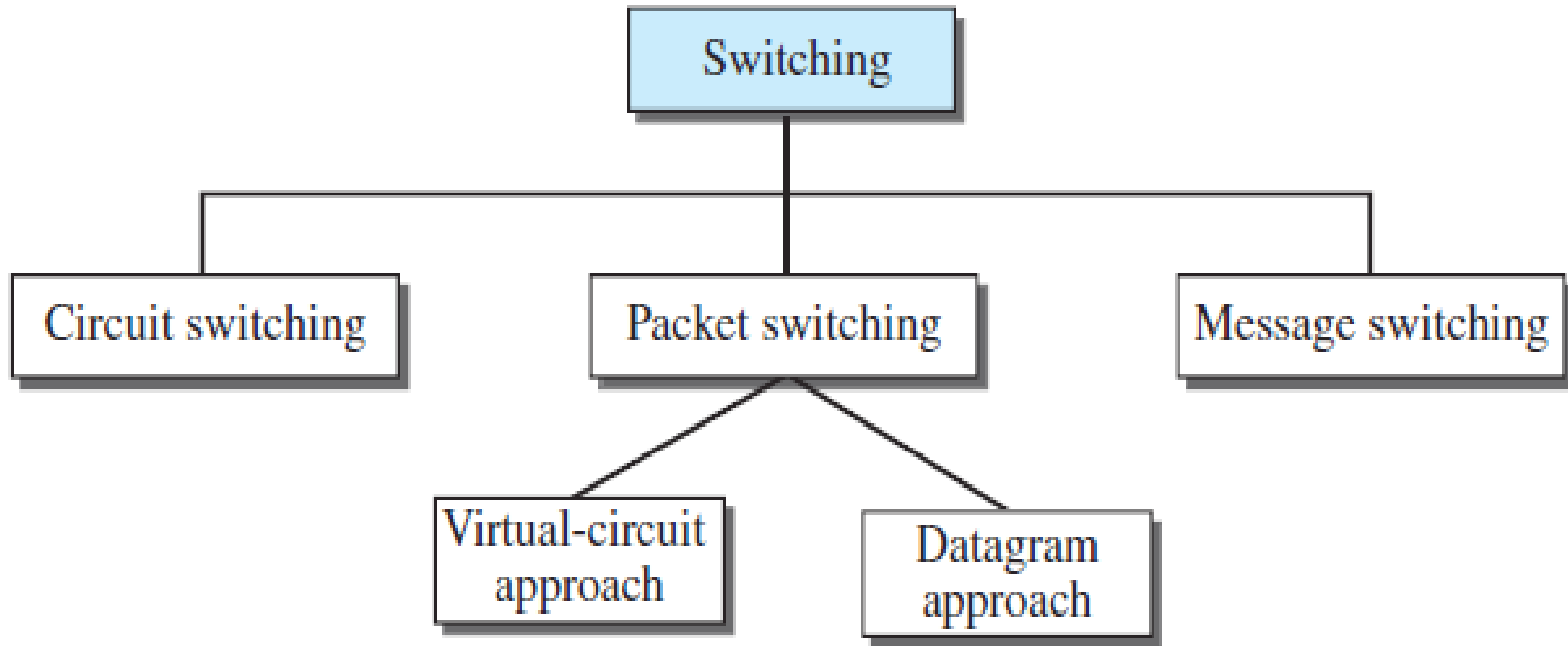
Introduction



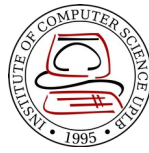
- Consists of a series of interlinked nodes called switches
- Switches are devices capable of creating temporary connections between two or more devices linked to the switch



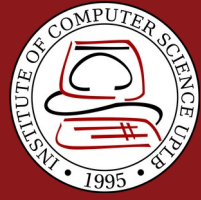
Switched Network



Switching and TCP/IP Layers

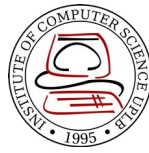


- Physical Layer
 - Circuit switching
- Data Link Layer
 - Packet switching (virtual approach)
- Network Layer
 - Packet switching (virtual or datagram approach)
- Application Layer
 - Message switching



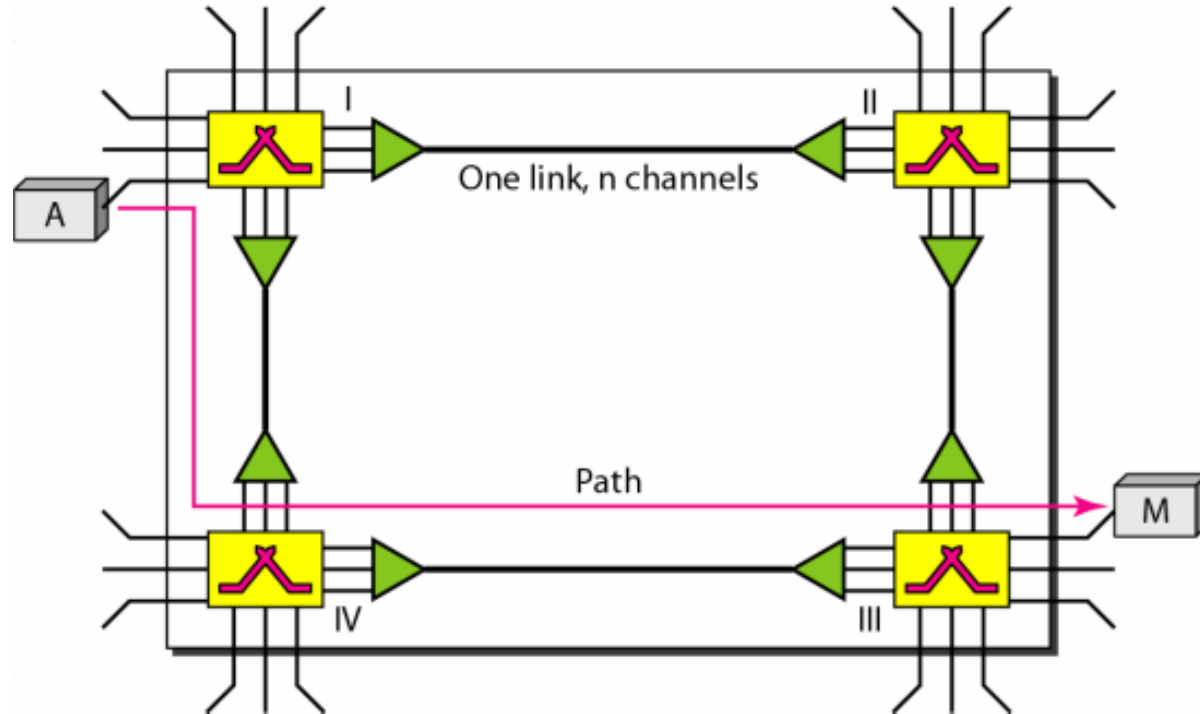
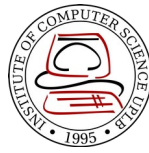
Circuit-Switched Network

Circuit-Switched Network

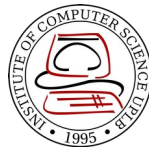


- Set of switches connected by physical links
- Connection between two stations is a dedicated path made of one or more links
- Only **one** channel in a link is used
- Each link is normally divided into n channels by FDM or TDM
- Phases: **connection setup**, **data transfer**, **connection teardown**
- Inefficient – dedicated link can be unused
- Minimal delay

Circuit-Switched Network



Circuit-Switched Network



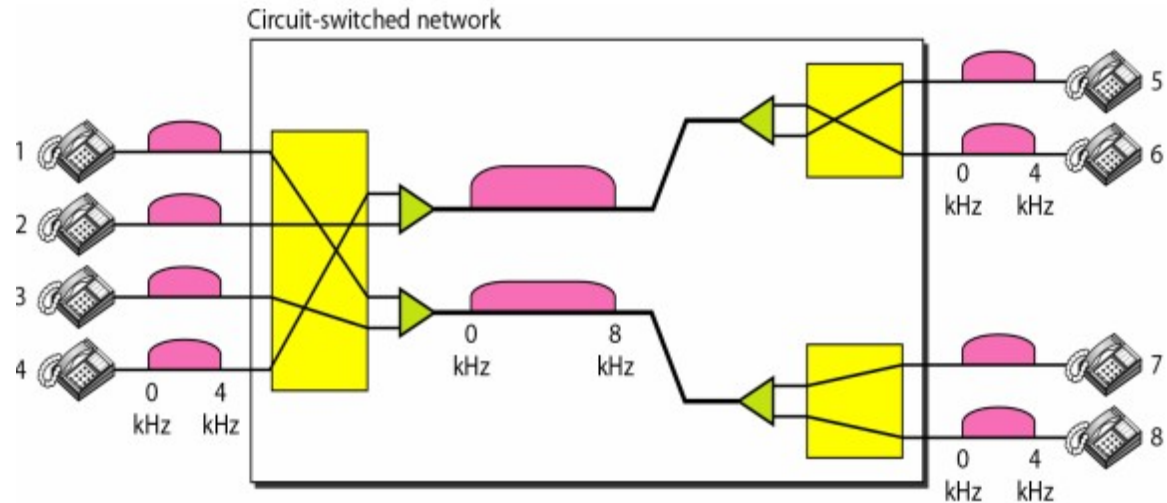
- Takes place at the **physical layer**
- Resources to be used must be reserved
 - bandwidth in FDM and time slot in TDM
 - Switch buffer
 - Switch processing time
 - Switch I/O ports
- Data transferred are a **continuous flow**
- **No addressing** involved in the data transfer

Circuit-Switched Network

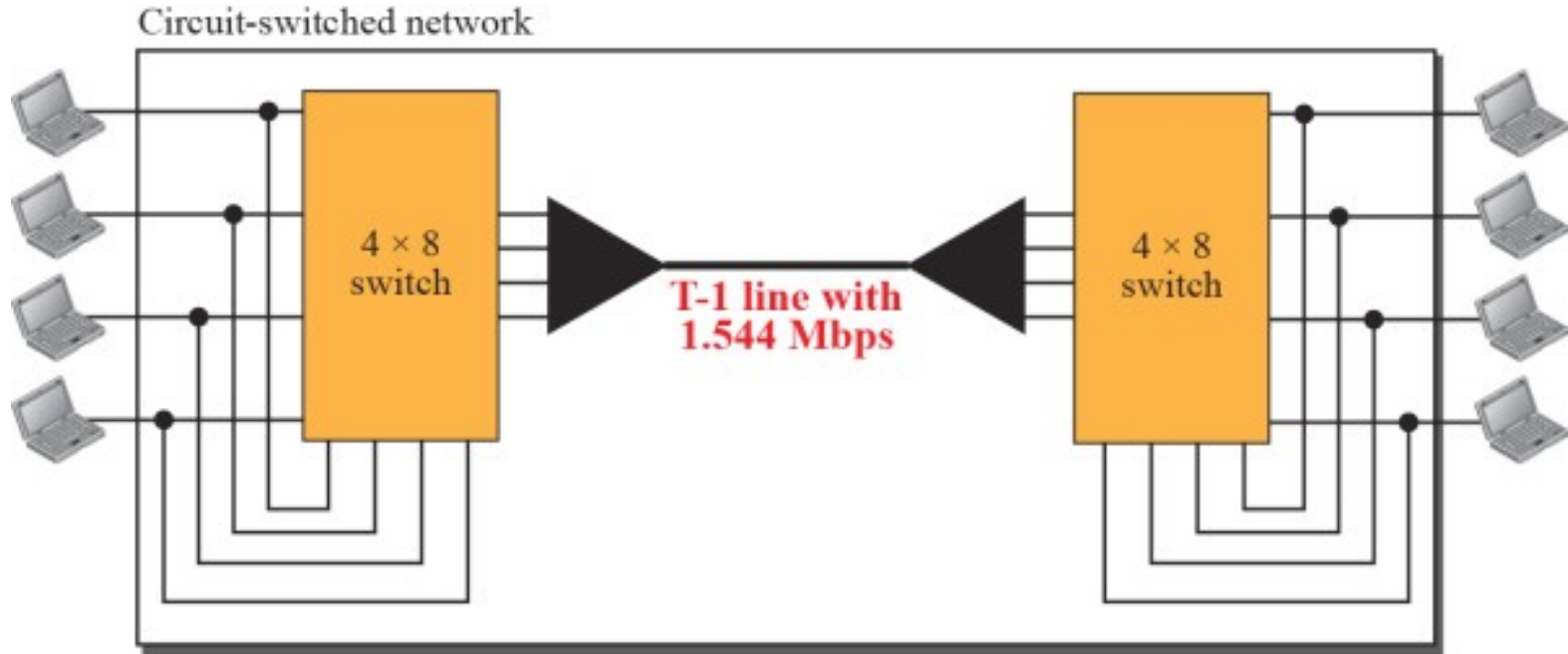


Cordboard

<http://ourwebhome.com/>



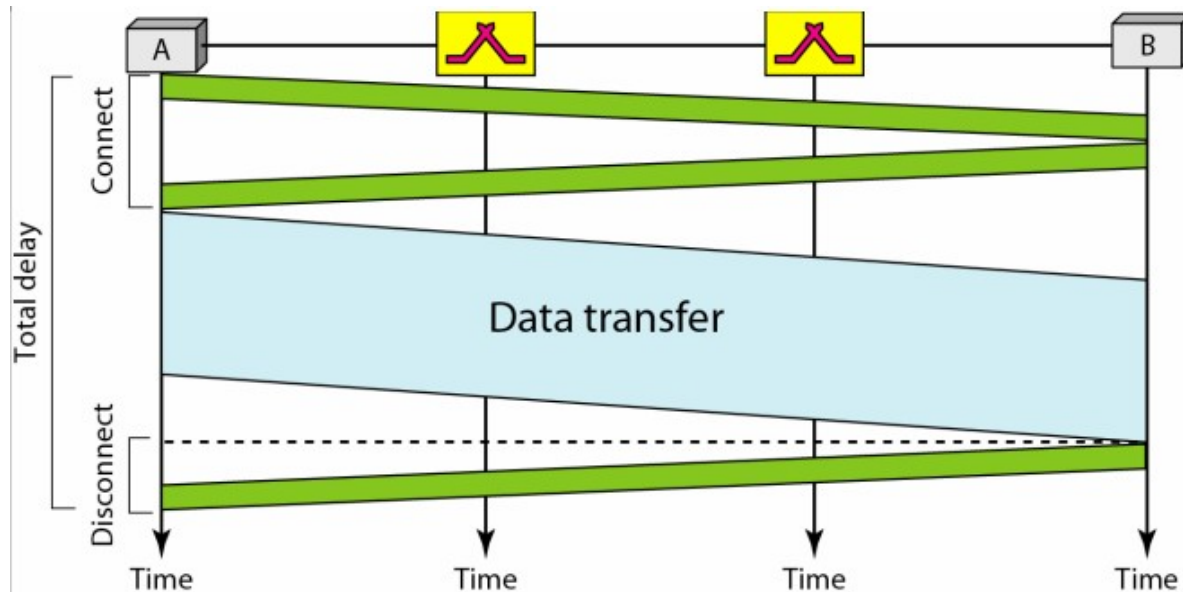
Circuit-Switched Network

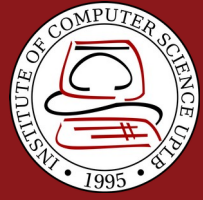


Circuit-Switched Network



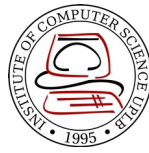
Delay is minimal – during data transfer, the data are not delayed at each switch; the resources are allocated for the duration of the connection





Packet Switched Network

Packet Switching

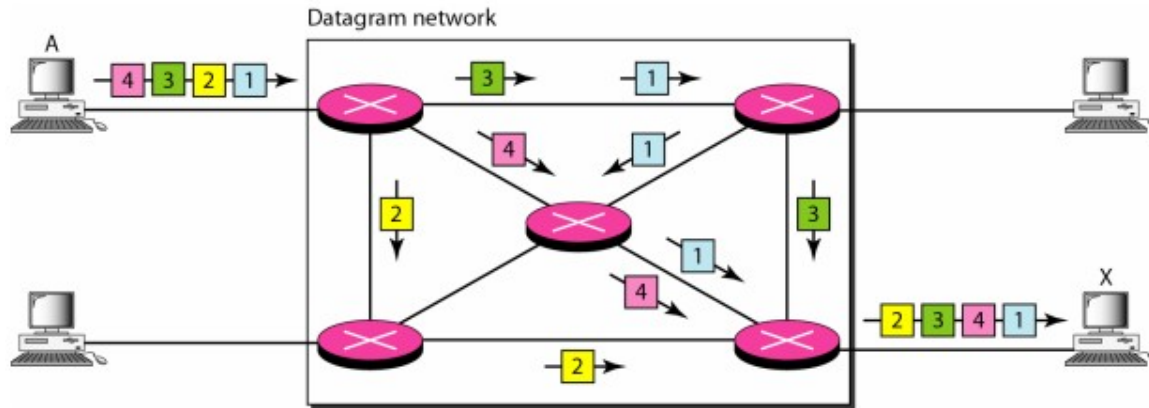


- In packet-switched networks, messages are divided into **packets**: *fixed* or *variable* size determined by the network and the governing protocol
- No resource allocation for a packet
- Resources are allocated on demand; FCFS
- 2 types: **datagram networks** and **virtual-circuit network**

Datagram Network



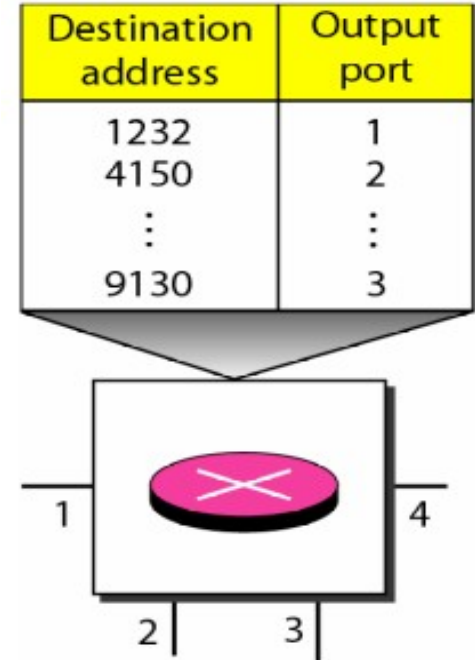
- Each packet is treated independently of all others
- Datagram switching done at the network layer
- Datagrams may arrive out of order or get lost
- Connectionless – no information on state of connection is saved



Datagram Network



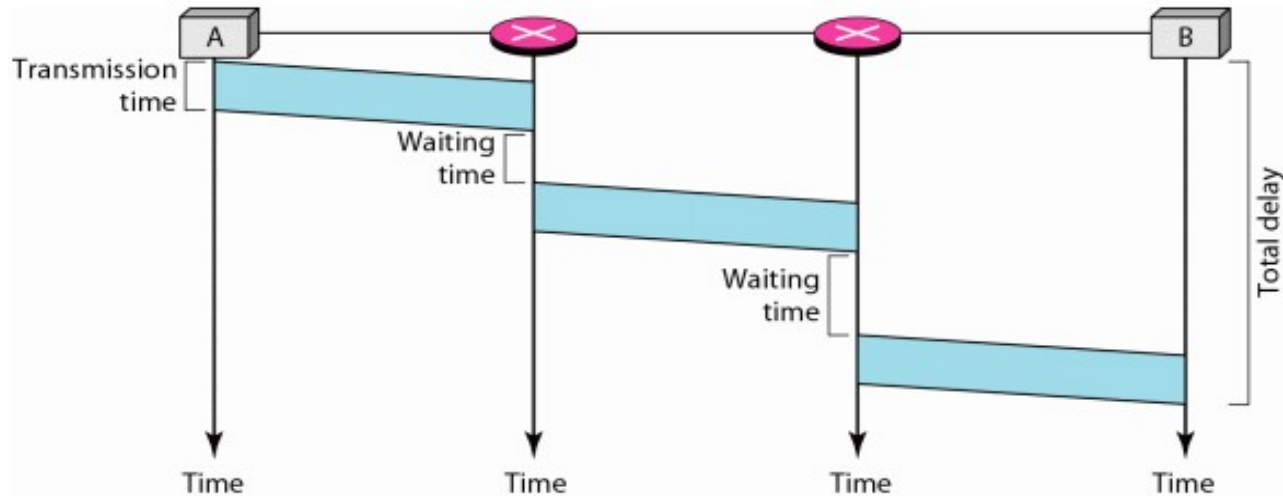
- A switch in a datagram network uses a routing table that is based on the destination address
- Routing tables are dynamic and are updated periodically
- Header contains address
- The destination address in the header of a packet in a datagram network remains the same during the entire journey of the packet



Datagram Network

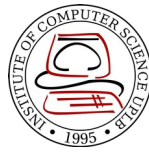


- Efficient – resource are allocated when there are packets to be transferred
- More delay – waiting time in switches; also since not all packets in a message travel through the same switches, delay is not uniform

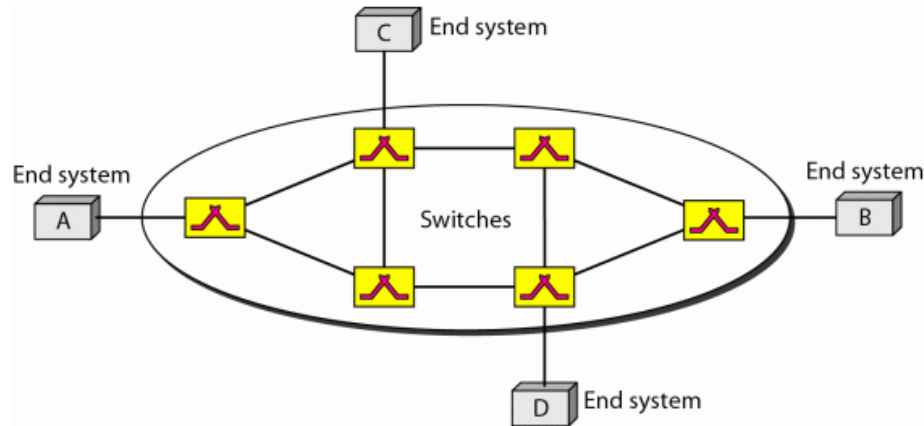


$$\text{Total Delay} = 3 \times \text{Transmission Time} + 3 \times \text{Propagation Time} + 2 \times \text{Waiting Time}$$

Virtual Circuit Network



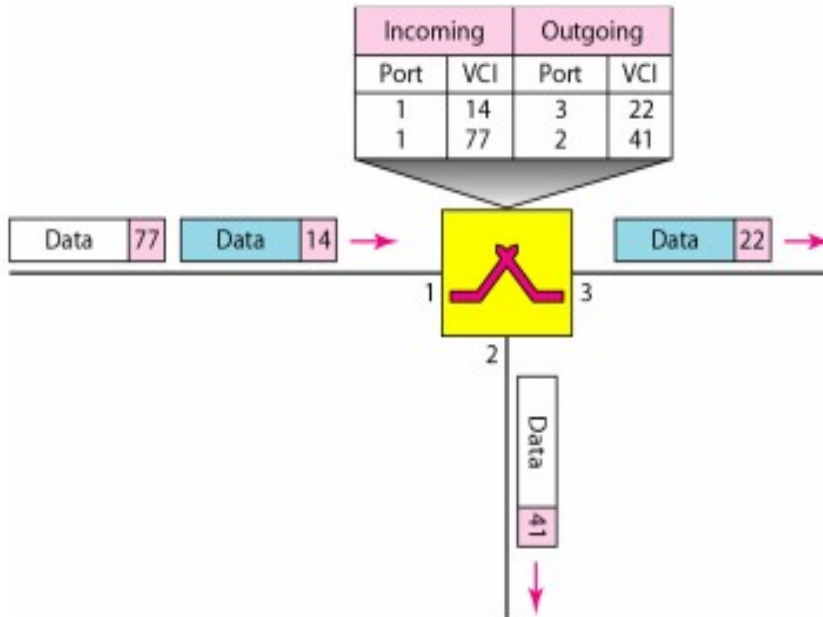
- Cross between circuit-switched network and datagram network
 - With connection setup and teardown phases
 - Resources can be allocated during the setup phase or on demand
 - Packetized data with address in header (local jurisdiction only)
 - Packets follow the same path established
 - Implemented in the data link layer



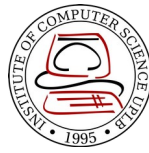
Virtual Circuit Network



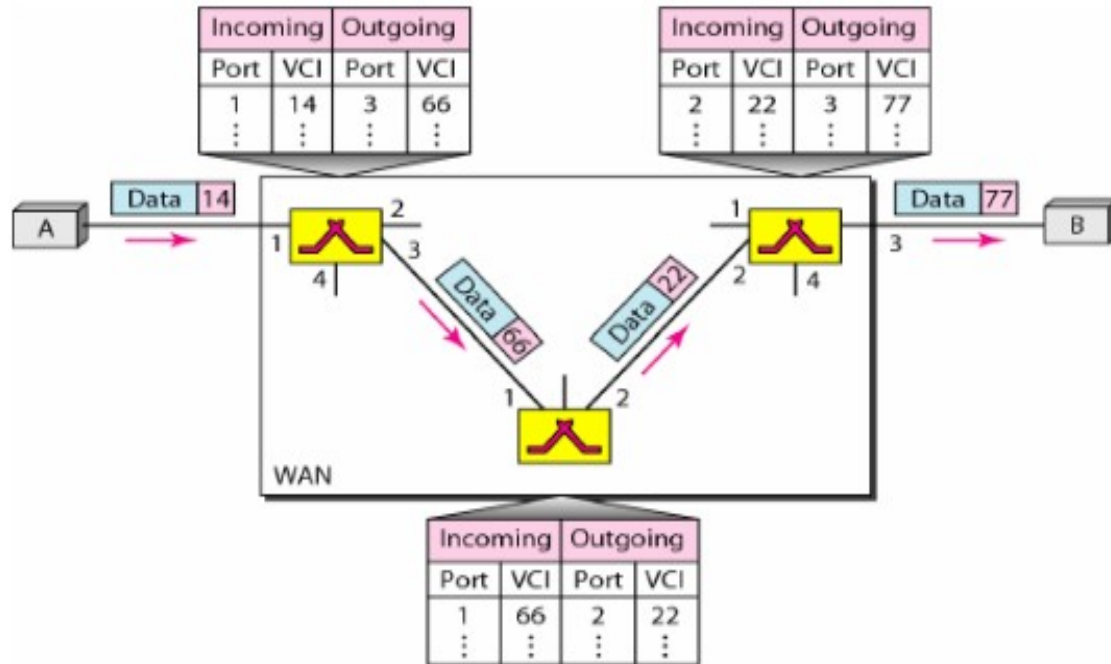
- A source or a destination needs to have a global address – used to create VCI
- **Virtual-Circuit Identifier** (VCI) – small number that has only switch scope (between two switches)



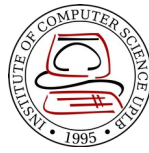
Virtual Circuit Network



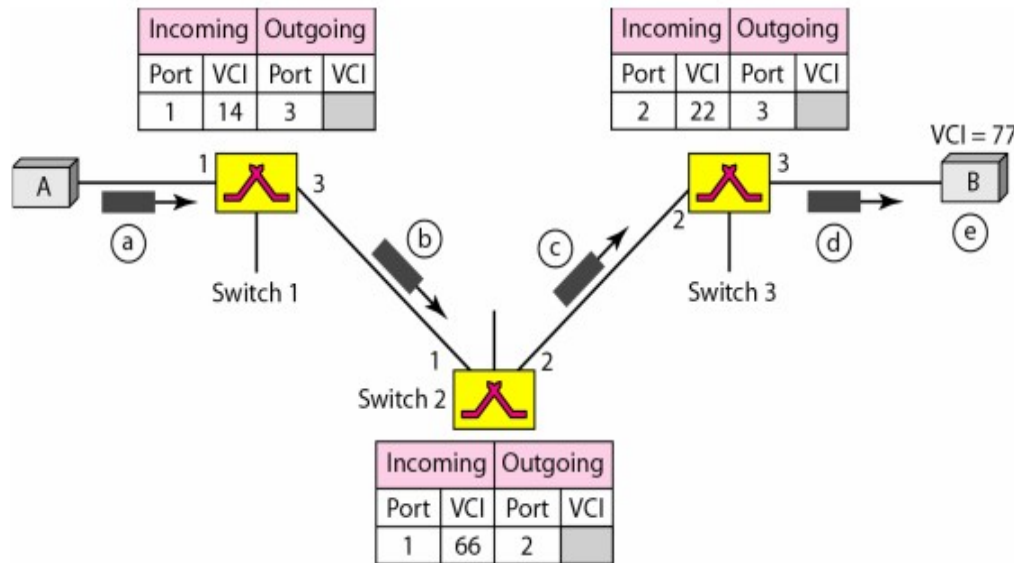
Data Transfer Phase – all switches need to have a table entry (in simplest form has 4 columns)



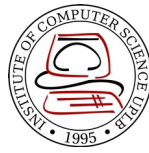
Virtual Circuit Network



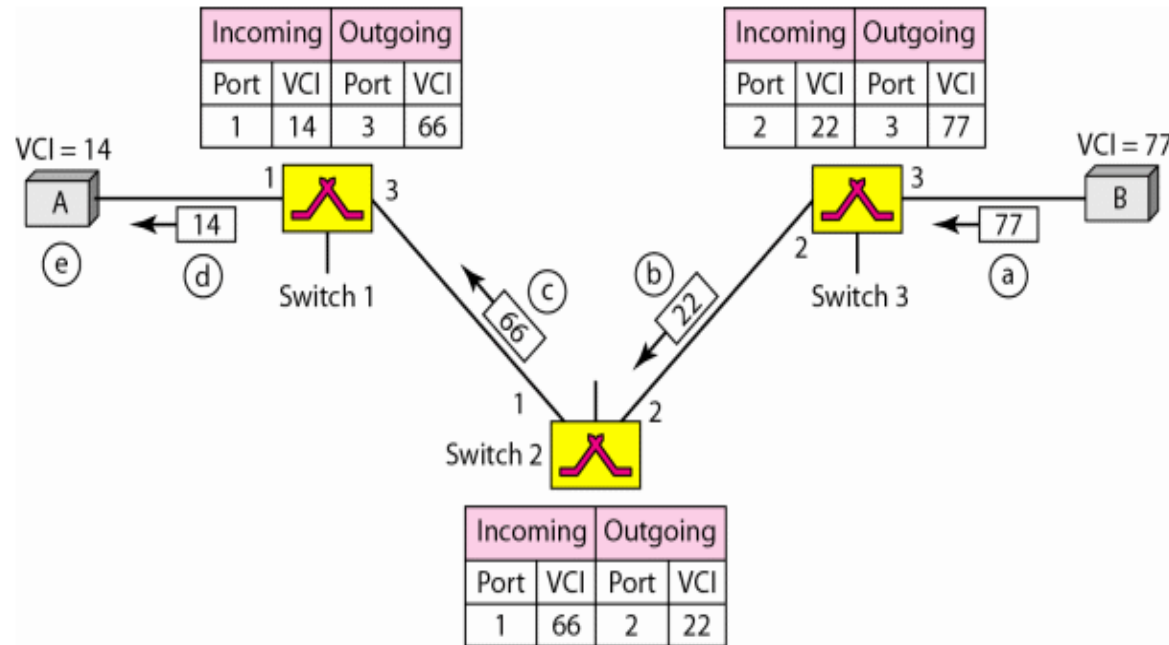
- In Setup phase, there are 2 required steps:
- Setup request – set-up request frame is sent from the source to the destination; switch acts as a packet switch (routing table is different from switching table)



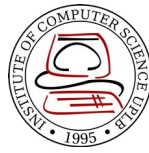
Virtual Circuit Network



- Acknowledgment - special frame (**acknowledgment frame**) completes the entries in the switching tables



Virtual Circuit Network

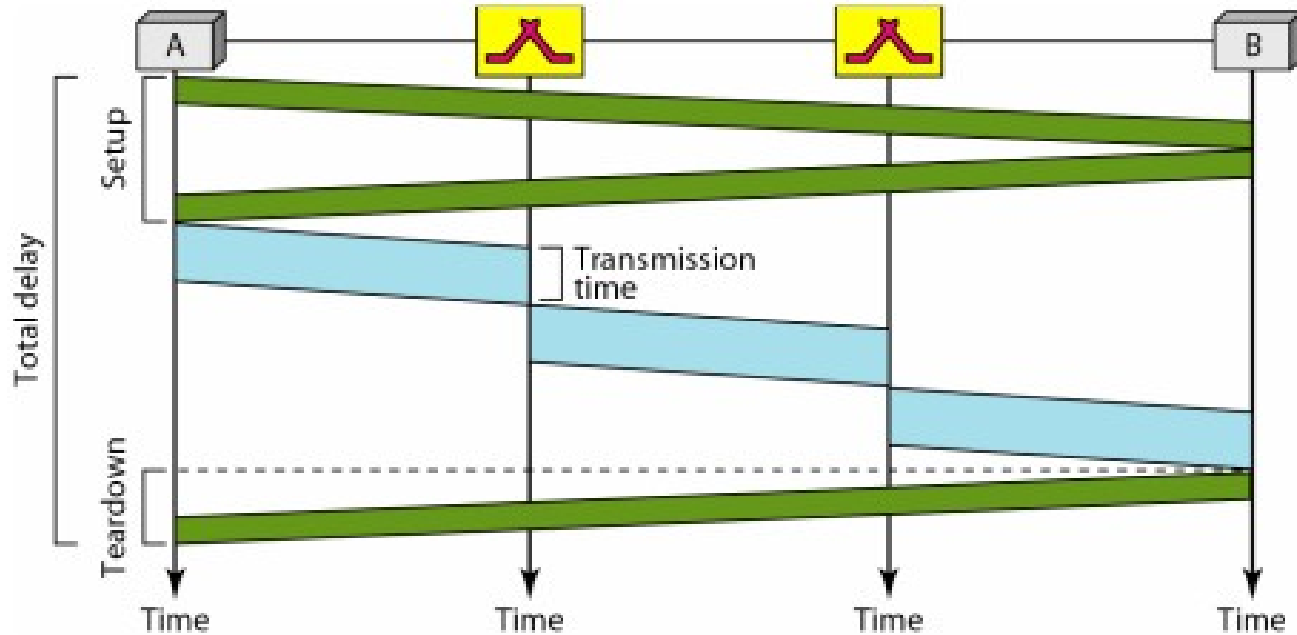


- Efficiency
 - All packets belonging to the same source and destination travel the same path but may arrive at the destination with different delays if resource allocation is on demand
 - If resource allocation is on demand, the source can check the availability of the sources without actually reserving it

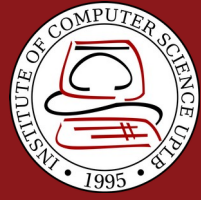
Virtual Circuit Network



Delay – there is one time delay for setup and one time delay for teardown

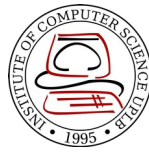


$$\text{Total Delay} = 3 \times \text{Transmission Time} + 3 \times \text{Propagation Time} + \text{Setup} + \text{Teardown}$$



Structure of a Switch

Structure of a Circuit Switch

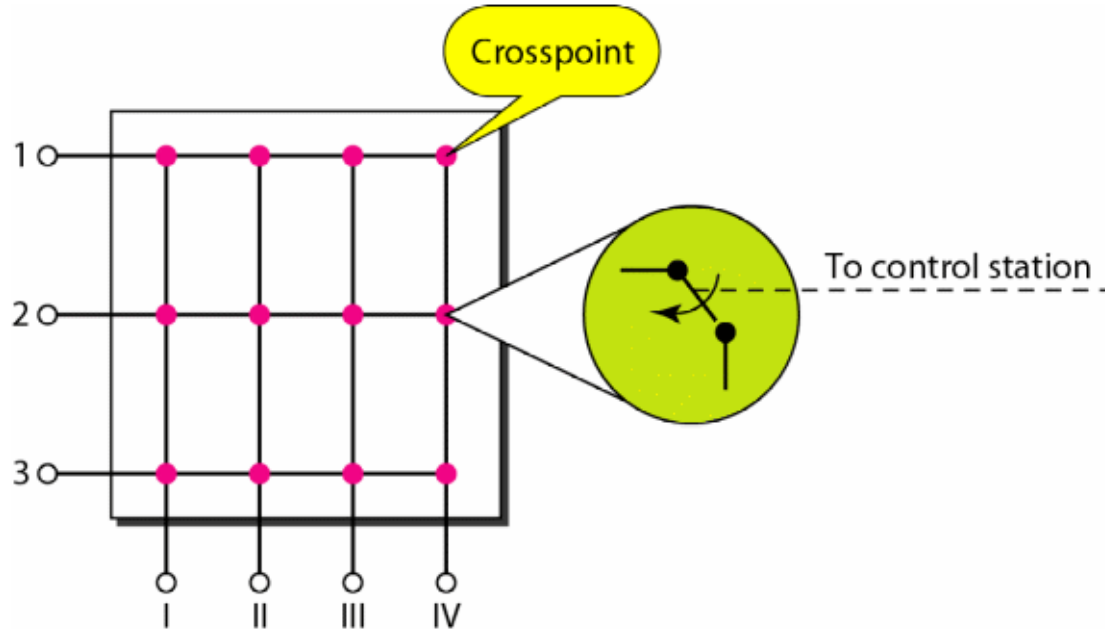


- **Space-division switch** – paths in the circuit are separated spatially
 - Crossbar (XBar)
 - Multistage
- **Time-division switch** – uses TDM
 - Time-slot interchange (TSI)
- Time- and Space-Division switch combo
 - Time-space-time

Structure of a Circuit Switch



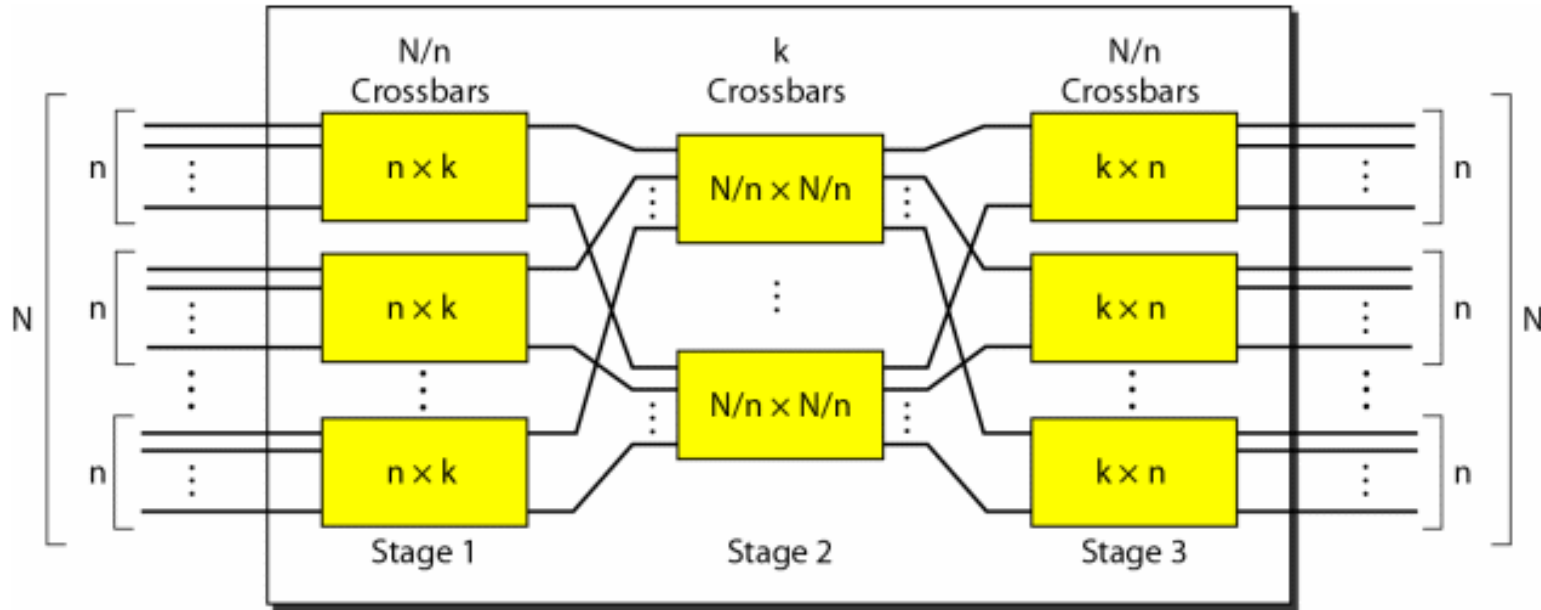
Crossbar switch – connects n inputs to m outputs in a grid using electronic microswitches (transistors) at each crosspoint



Structure of a Circuit Switch



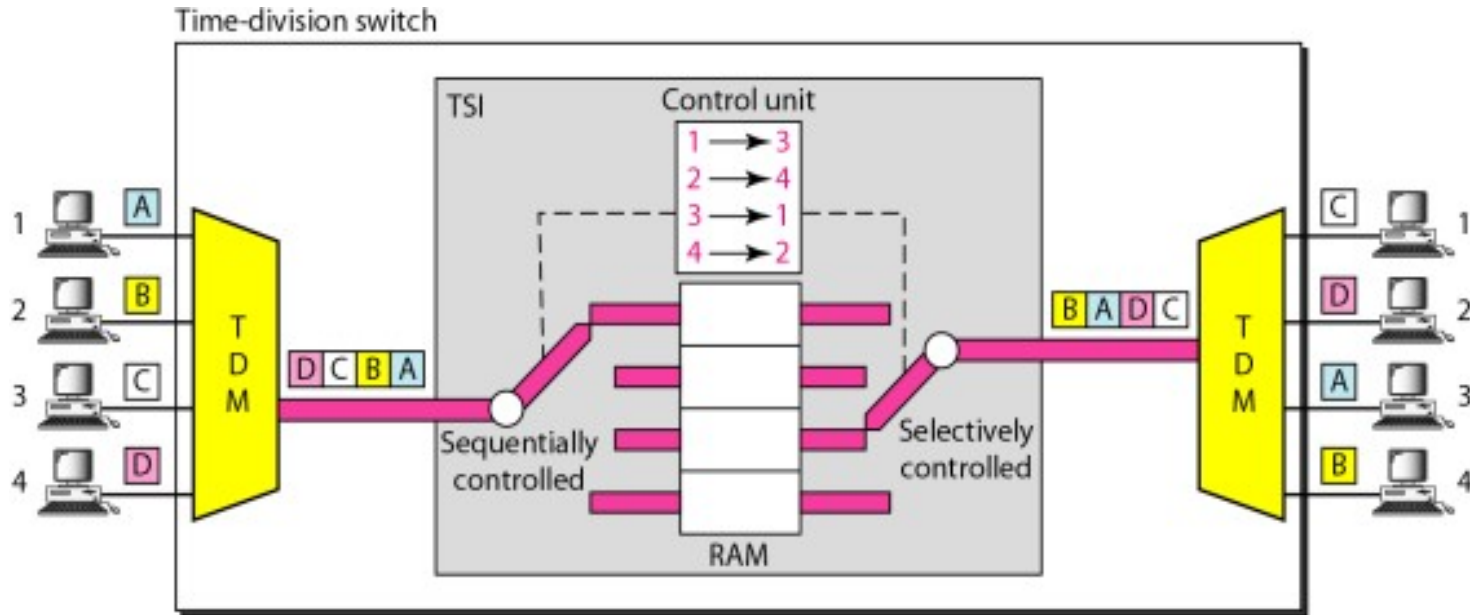
Multistage switch – combines crossbar switches in several stages



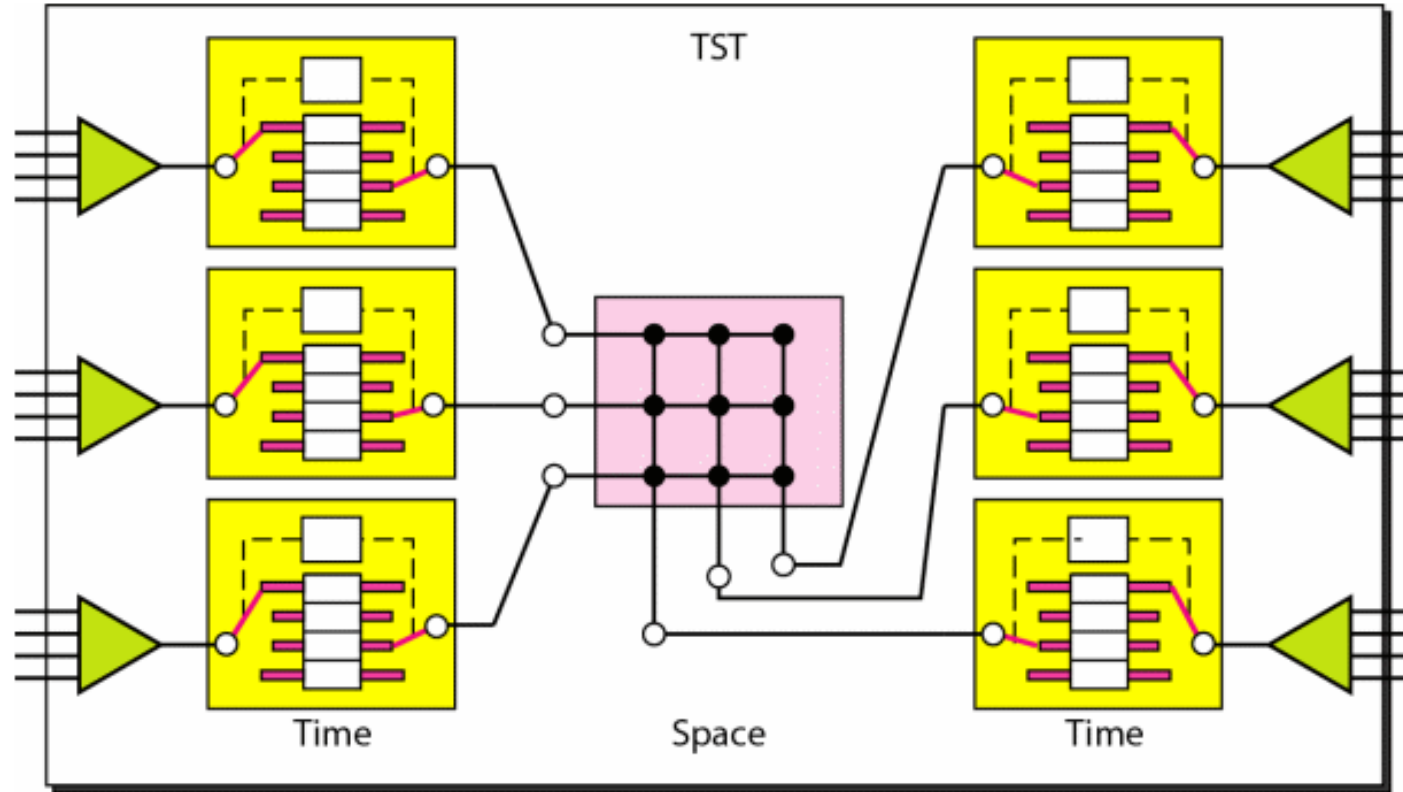
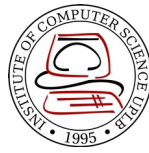
Structure of a Circuit Switch



Time-slot Interchange (TSI) – consists of random access memory (RAM) with several memory locations



Structure of a Circuit Switch



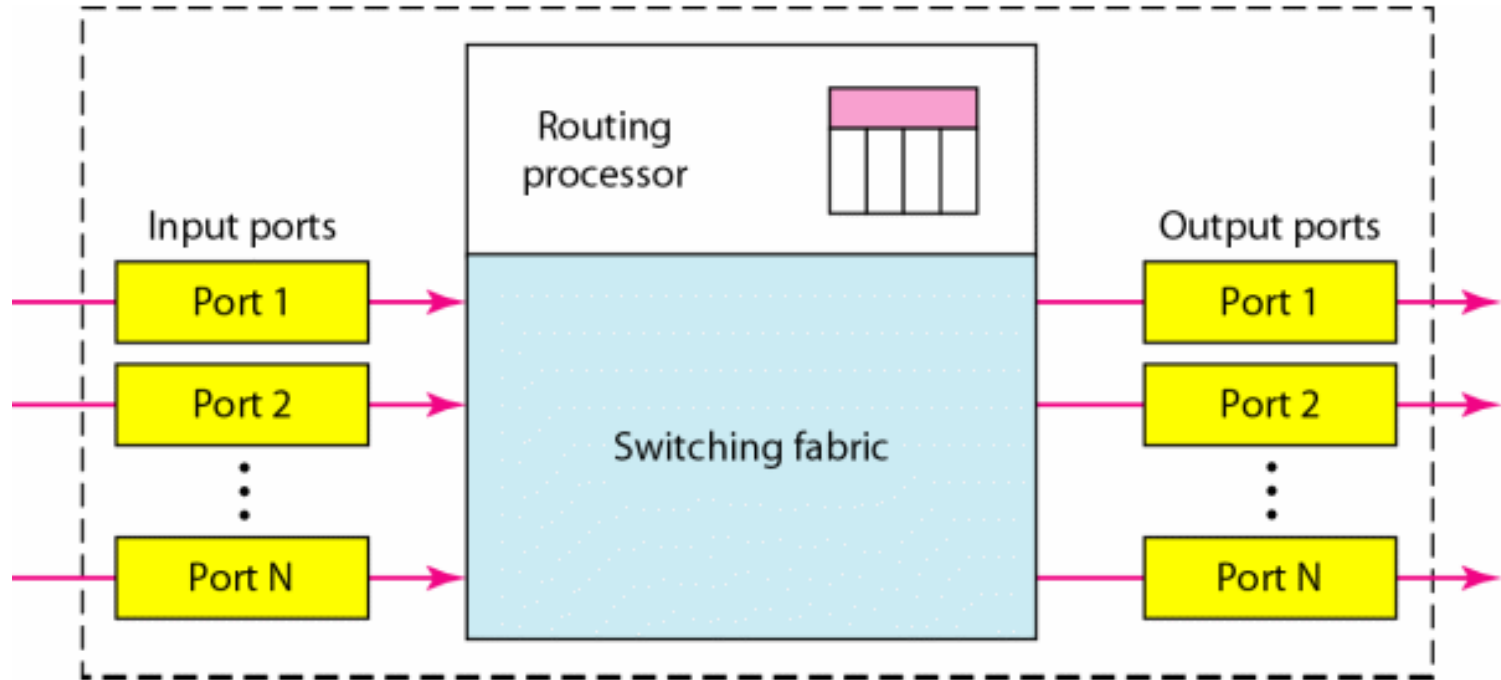
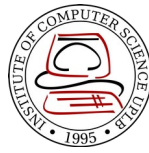
Time-space-time switch

Structure of a Packet Switch

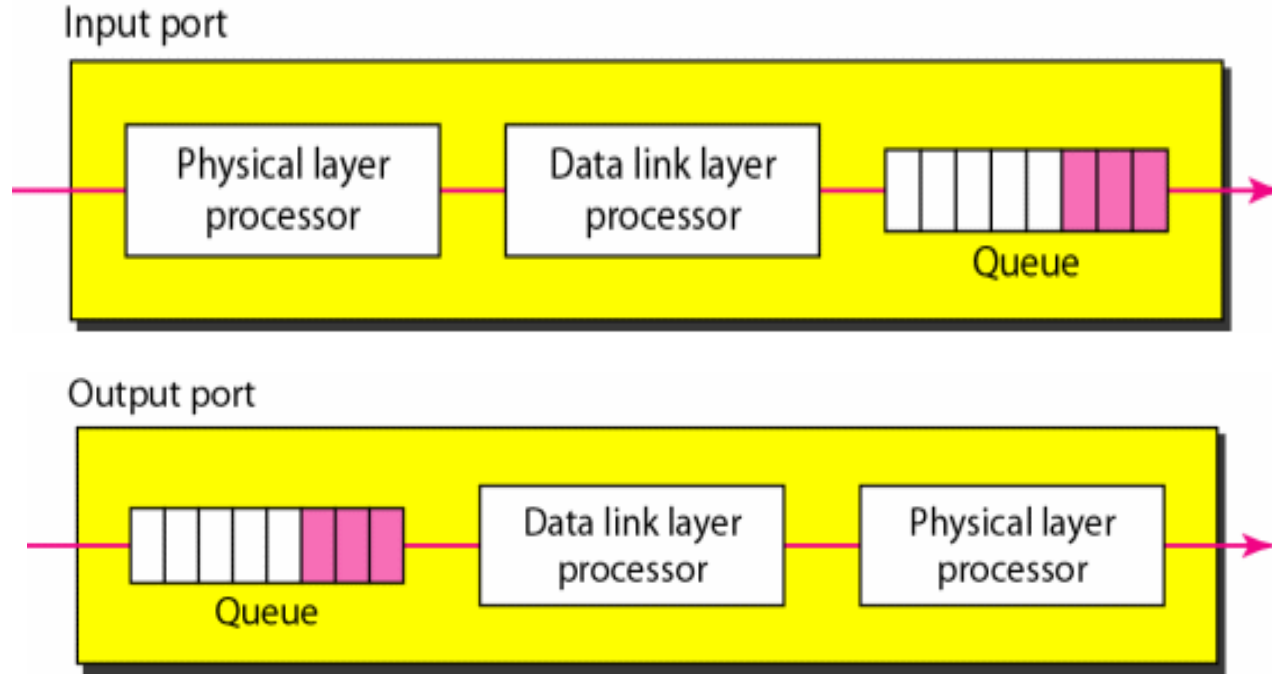
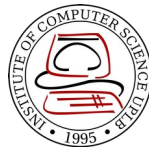


- Four components
 - **Input ports** – performs the physical and data-link functions of the packet switch
 - **Output ports** – performs same function as input ports in reverse order
 - **Routing processor** – perform function of the network layer
 - **Switching fabric**
 - Crossbar
 - Banyan
 - Batcher-Banyan Switch

Structure of a Packet Switch



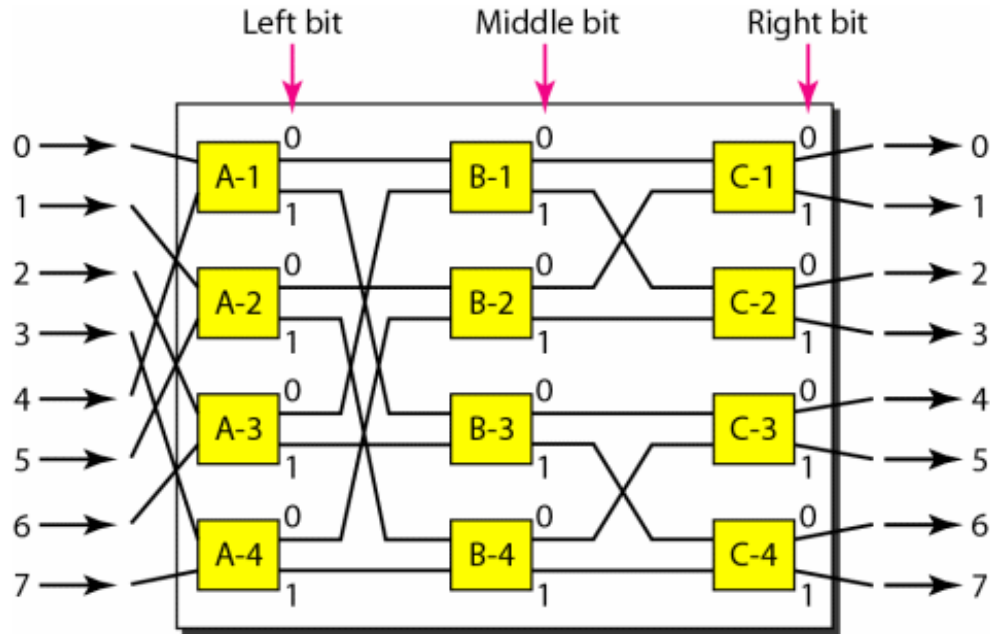
Structure of a Packet Switch



Structure of a Packet Switch



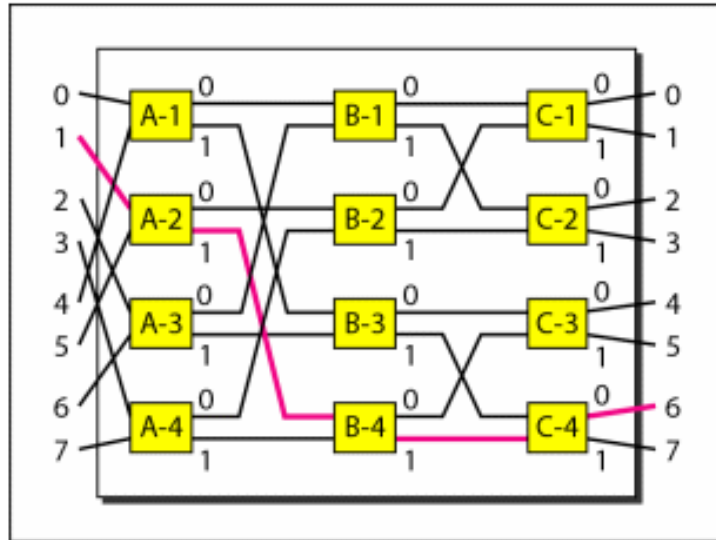
Banyan switch – multistage switch with microswitches at each stage that route the packets based on the output port represented as a binary string



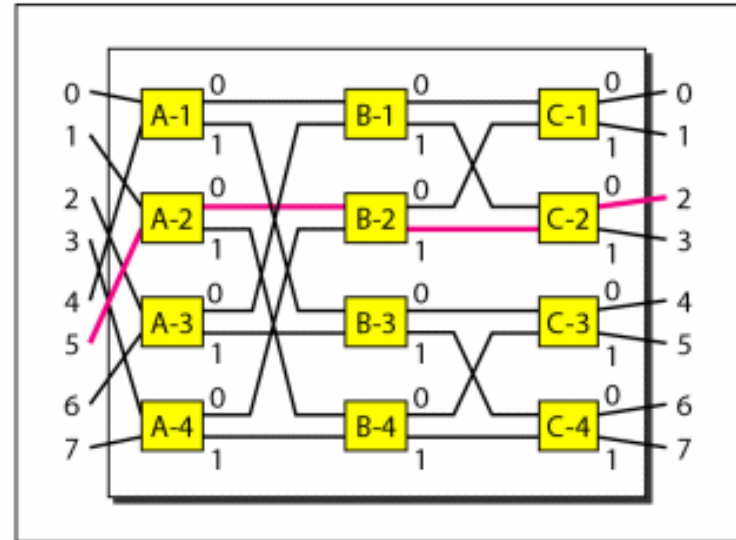
Structure of a Packet Switch



Routing in a Banyan Switch



a. Input 1 sending a cell to output 6 (110)



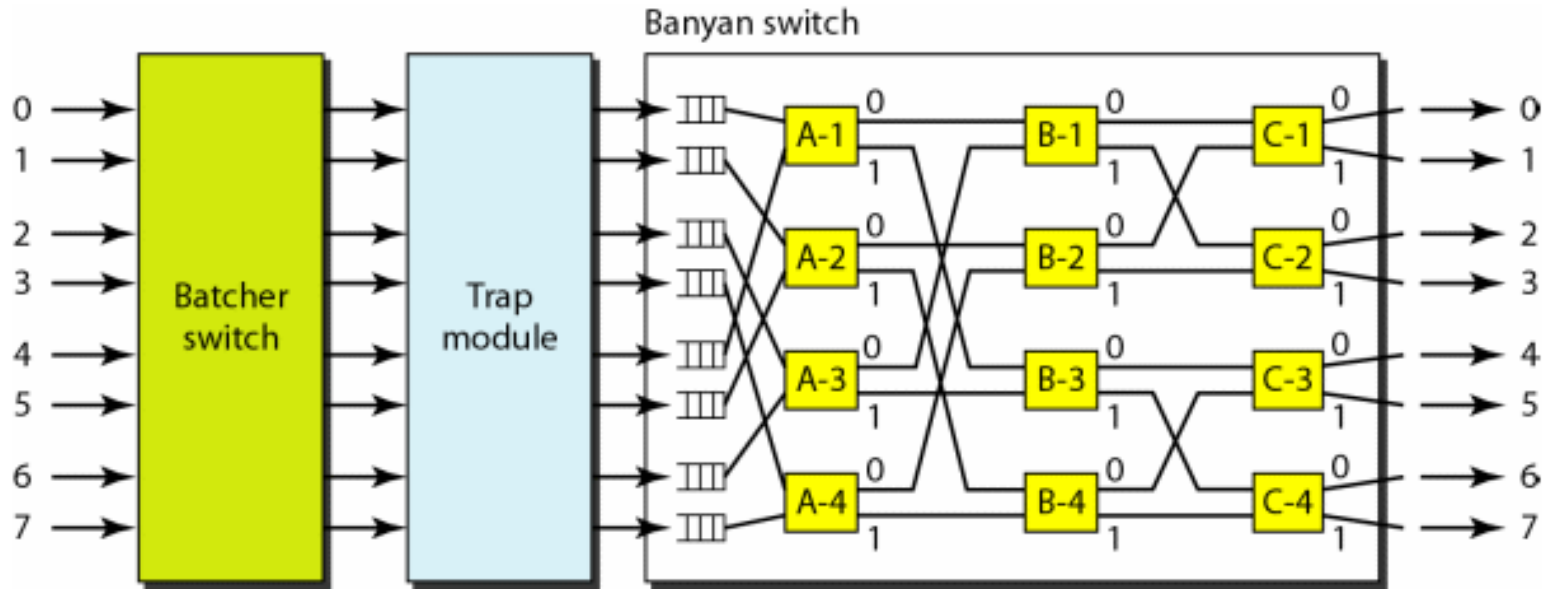
b. Input 5 sending a cell to output 2 (010)

Problem: Possibility of internal collision even when two packets are not heading for the same output port

Structure of a Packet Switch



Batcher-Banyan Switch – combination of Banyan switch and a switch designed by K.E. Batcher which sorts incoming packets according to their final destinations



References



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.